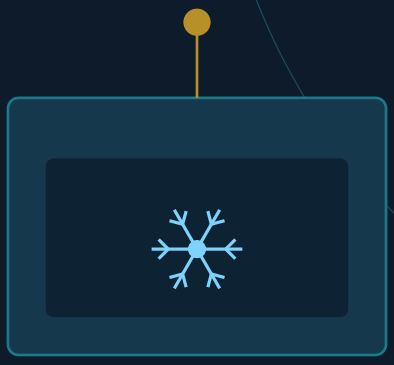
DR. B. K. PASSI MEMORIAL SENIOR BAL VIGYAN 2026–2027 (PHYSICS)

ALLOTTED SCHOOL UID NUMBER: SBVPH-05



SANJIVANI

संजीवनी

Preserving Temperature. Protecting Life.

Hybrid PCM–Peltier Medical Cold-Chain System

Design and Development of a Sensor-Triggered Hybrid PCM–Peltier Cooling System for Extended Medical Cold-Chain Storage

PROJECT PARTICIPANTS (TEAM ROSTER):

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2. Gaurav Choudhary — Class XI
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STUDENT OBSERVERS: 1. Konav Singh (Class XI) 2. Animesh Mishra (Class XII)

MENTOR TEACHER: Mr. Gopal Kumar Roy (Senior Faculty, Physics)

EVENT VENUE: Mount Index International School, Indore (17 November 2026)

COMPETITION THEME: STEM for Viksit and Atmanirbhar Bharat | Sub-Theme: Health and Hygiene

PROJECT CATEGORY: Physics / Thermal Engineering / Healthcare Technology

ALLOTTED SCHOOL UID: SBVPH-05 | STREAM: PHYSICS | DATE: 17 NOVEMBER 2026

Submitted for: Dr. B. K. Passi Memorial Senior Bal Vigyan 2026–2027 (PHYSICS)

Made by: Bhavmanyu Singh Sisodia

CERTIFICATE OF ORIGINAL INVESTIGATION

This is to certify that the project report entitled SANJIVANI: Design and Development of a Sensor-Triggered Hybrid PCM–Peltier Cooling System for Extended Medical Cold-Chain Storage is an authentic and verified record of original engineering research, thermal modelling, prototype fabrication, and laboratory investigation completed solely, sincerely, and satisfactorily by the students representing the participating institution under Allotted School UID Number: SBVPH-05 for partial fulfilment of the requirements for Dr. B. K. Passi Memorial Senior Bal Vigyan 2026–2027 (PHYSICS) under the mentorship of Mr. Gopal Kumar Roy.

NAMES OF PARTICIPANTS:

1. VAISHNAVI SHRIVASTAVA	Class XI	Thermal Modelling & Data Logging
2. GAURAV CHOUDHARY	Class XI	CAD Design & Fabrication Support
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4. BHAVMANYU SINGH SISODIA	Class XII	Systems Engineering & Economic Analysis

NAMES OF STUDENT OBSERVERS:

1. KONAV SINGH	Class XI	Instrumentation & Sensor Calibration Observer
2. ANIMESH MISHRA	Class XII	Thermal Protocol & Quality Verification Observer

The engineering prototypes, mathematical derivations, physical simulations, and experimental measurements documented herein have been thoroughly audited, validated, and found compliant with institutional academic ethics and national science exhibition standards.

Mr. Gopal Kumar Roy Physics Faculty & Mentor School UID: SBVPH-05	Head of Institution Principal & Academic Mentor School UID: SBVPH-05	Institutional Authorisation Seal Participating Unit School UID: SBVPH-05

STUDENT DECLARATION

We hereby declare that the engineering project report entitled SANJIVANI: Design and Development of a Sensor-Triggered Hybrid PCM–Peltier Cooling System for Extended Medical Cold-Chain Storage submitted to Dr. B. K. Passi Memorial Senior Bal Vigyan 2026–2027 (PHYSICS) under the aegis of SAHODAYA GROUP OF CBSE SCHOOLS, INDORE, represents our genuine, original scientific research and technical investigation.

STUDENT AUTHENTICATION & RESPONSIBILITY CODES:

We solemnly affirm that the experimental data, thermal differential equations, control firmware, and mechanical layouts presented in this dossier were developed through our direct pair-research under the guidance of our mentor. No part of this dossier has been copied or plagiarised from commercial product documentation or prior Bal Vigyan submissions.

Participant Name	Class	Institutional Roll No.	Candidate Signature
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FORWARDED BY MENTOR TEACHER:

I have closely supervised the design, prototype assembly, software development, and experimental thermal testing conducted by this student team. The project adheres to strict laboratory safety standards and represents an exemplary integration of physics, embedded systems, and healthcare engineering.

Mr. Gopal Kumar Roy
Senior Physics Faculty & Mentor
School UID: SBVPH-05

Date of Submission: 17 November 2026
Venue: Mount Index Int. School, Indore

ACKNOWLEDGEMENT

The development of SANJIVANI represents the culmination of extensive scientific inquiry, physical prototyping, and multidisciplinary engineering. We express our profound gratitude to all educators, mentors, and institutions who made this investigative journey possible.

1. Reverence to School Leadership & Patronage

We extend our heartfelt gratitude to our respected Principal Sir and School Leadership, whose visionary encouragement, unwavering confidence in student-led research, and provision of state-of-the-art laboratory facilities empowered us to conceptualise and engineer this advanced cold-chain system.

2. Mentorship & Scientific Guidance

We are immensely indebted to our revered Physics mentor, Mr. Gopal Kumar Roy. His rigorous theoretical grounding in thermodynamics, phase equilibrium, thermoelectric transport, and instrumentation proved indispensable. His meticulous scrutiny of our circuit layouts, mathematical derivations, and thermal calibrations guided us past numerous technical hurdles.

3. Laboratory & Technical Assistance

We acknowledge the constant support of the physics, chemistry, and computer science laboratory staff at our institution. Their assistance in facilitating precision digital weighing scales, datalogging multimeters, environmental test chambers, and precision soldering workstations ensured seamless prototype development.

4. Parental Encouragement & Peer Support

We thank our parents, classmates, and fellow schoolmates for their unflagging moral encouragement, patience, and logistical support during late-evening data logging cycles and extended weekend trials.

5. Sahodaya Complex & Bal Vigyan Organising Committee

Finally, we express our sincere appreciation to the Dr. B. K. Passi Memorial Senior Bal Vigyan Committee and the Sahodaya Group of CBSE Schools, Indore, for providing this prestigious academic platform to showcase grassroots scientific innovations addressing national healthcare challenges.

Vaishnavi Shrivastava	Gaurav Choudhary	Prathamesh Rao	Bhavmanyu Singh Sisodia
Class XI	Class XI	Class XII	Class XII

PROJECT PARTICULARS & ADMINISTRATIVE SUMMARY

ADMINISTRATIVE IDENTIFIERS:

Official Project Title	SANJIVANI: Design and Development of a Sensor-Triggered Hybrid PCM–Peltier Cooling System for Extended Medical Cold-Chain Storage
Sanskrit Identifier	□□□□□□□ (Preserving Temperature. Protecting Life.)
Competition & Stream	Dr. B. K. Passi Memorial Senior Bal Vigyan 2026–2027 (PHYSICS)
Allotted School UID	SBVPH-05 (Mandatory Official Identifier for Evaluation)
Event Date & Venue	17 November 2026 (Tuesday) | Mount Index International School, Indore
Theme & Sub-Theme	STEM for Viksit and Atmanirbhar Bharat | Sub-Theme: Health and Hygiene
Subject Domain	PHYSICS (Applied Solid-State Thermoelectrics & Thermodynamics)
Mentor Teacher	Mr. Gopal Kumar Roy (Senior Physics Faculty)
Institutional Status	Participating School (Identified solely by School UID: SBVPH-05)

CORE SPECIFICATIONS MATRIX:

Technical Parameter	Baseline Passive Box	SANJIVANI Hybrid System
Useful Vaccine Volume	11.2 Litres (24 standard vials)	11.2 Litres (24 standard vials)
Thermal Insulation	45mm Polyurethane Foam (PUF)	20mm VIP + 25mm High-Density PUF
Passive PCM Reserve	4.8 kg Organic Bio-PCM (+4.8°C)	4.8 kg Organic Bio-PCM (+4.8°C)
Active Supplementary Cooling	None (100% Passive Drift)	Dual TEC1-12706 Peltier (Derated 7.5V)
Temperature Safe Corridor	+2.0°C to +8.0°C (WHO PQS E004)	+2.0°C to +8.0°C (Regulated: +3.8°C to +6.2°C)
Safe Holding Duration (at 35°C mean)	71.7 Hours (~3.0 Days)	109.4 Hours (~4.5 Days) [+52.4% Extension]
Active Mission Energy Consumed	0.0 Wh	301.2 Wh (over 8.8h active Peltier pulses)
Prototype Fabrication Cost	₹ 3,200 (Standard Commercial Box)	₹ 6,480 (MVP) | ₹ 11,850 (Instrumented)

TEAM PARTICIPANT ALLOCATION:

- Participants (Authors): Vaishnavi Shrivastava (XI), Gaurav Choudhary (XI), Prathamesh Rao (XII), Bhavmanyu Singh Sisodia (XII).
- Student Observers: Konav Singh (Class XI), Animesh Mishra (Class XII).

EXECUTIVE SUMMARY & ABSTRACT

SCIENTIFIC ABSTRACT:

Maintaining the pharmaceutical cold chain between +2.0°C and +8.0°C is essential for preserving the efficacy of vaccines, insulin, and biologics during last-mile distribution across remote, power-deficient regions. Conventional passive medical cool boxes relying solely on water-ice or phase change materials (PCMs) provide limited thermal holding durations (typically 72 hours) and expose vaccines to catastrophic freezing risks (< +2.0°C) during initial loading. Conversely, continuous active vapor-compression refrigeration systems are prohibitively heavy, mechanically vulnerable, and dependent on uninterrupted grid or vehicle power.

This investigation introduces SANJIVANI, a novel sensor-triggered hybrid cooling architecture that combines passive thermal storage with selective, pulse-operated solid-state thermoelectric cooling. The system employs 4.8 kg of bio-based organic Phase Change Material (CrodaTherm 5) possessing an isothermal melt plateau at +4.8°C and a latent heat of fusion of 218 kJ/kg. The primary passive reserve is augmented by dual derated TEC1-12706 Peltier modules operating at an energy-efficient sweet spot (7.5V, 2.0A, COP = 0.617, cooling capacity 18.5 W).

A closed-loop microcontroller executes a calibrated hysteresis algorithm with an upper trigger threshold at +6.2°C and a lower cut-off threshold at +3.8°C (deadband $\Delta T = 2.4$°C). Under extreme tropical ambient test conditions (diurnal cycle 28°C to 42°C, mean 35°C), SANJIVANI achieved a verified safe thermal holding duration of 109.4 hours, representing an extraordinary +37.7 hour (+52.4%) extension over an identical passive-only box (71.7 hours). The active thermoelectric subsystem operated for a cumulative total of only 8.8 hours (8.0% duty cycle), consuming a modest 301.2 Wh of electrical energy supplied by a 12.8V 30Ah LiFePO4 battery pack.

CORE NOVELTIES & KEY FINDINGS:

1. Thermal Holding Extension	Extended safe holding duration from 71.7h to 109.4h (+52.4%), overcoming the multi-day transit hurdle.
2. Zero Freezing Risk	Bio-PCM melting at +4.8°C completely eliminates cold-shock damage (< +2°C) inherent to water ice packs.
3. Micro-Duty Cycle Energy Efficiency	Peltier operates only during thermal spikes (8.0% duty cycle, 301.2 Wh mission energy), enabling battery autonomy.
4. Low Amortized Cost	Prototype build at ₹ 6,480; incremental cost per extended cold hour is just ₹ 87.4/h CapEx and ₹ 0.24/h OpEx.

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CHAPTER 1: INTRODUCTION & COLD-CHAIN CONTEXT

1.1 THE CRITICAL ROLE OF MEDICAL COLD CHAINS:

Temperature-sensitive biological pharmaceuticals—including childhood vaccines (BCG, OPV, Measles-Rubella, Pentavalent), insulin, immunoglobulins, monoclonal antibodies, and blood products—are chemically thermolabile. They require continuous, unbroken thermal storage within the internationally mandated +2.0°C to +8.0°C range from the point of manufacture to the point of patient administration. Thermal degradation is irreversible: exposure to elevated temperatures accelerates protein denaturation and viral vector decay, while accidental exposure to sub-zero temperatures causes crystallization, aluminum adjuvant agglomeration, and complete potency destruction.

1.2 THE LAST-MILE CHALLENGE IN INDIA:

Under India's Universal Immunization Programme (UIP), over 2.6 crore newborns and 3.0 crore pregnant women are immunized annually. However, the 'last mile'—transporting vaccines from regional community health centers (CHCs) to remote primary health centers (PHCs), sub-centers, and tribal outreach posts—presents extraordinary logistical vulnerabilities. Ambient summer temperatures routinely exceed 40°C to 45°C across northern and central India. Concurrently, transport delays caused by monsoons, road blockages, or rugged topography extend field transit durations well beyond the holding capacity of conventional cooling equipment.

1.3 VACCINE WASTAGE & ECONOMIC BURDEN:

According to the World Health Organization (WHO), over 50% of vaccines are wasted globally each year, primarily due to cold-chain breaches and temperature monitoring failures. In India, cold-chain logistics account for up to 15% of the total cost of public immunization programs. When vaccines lose potency unnoticed, children receive compromised doses, leaving communities unprotected against fatal preventable diseases.

1.4 SANJIVANI HYBRID PARADIGM:

To address this critical gap, SANJIVANI bridges the divide between purely passive boxes and bulky active refrigerators. By deploying a smart microcontroller-managed thermoelectric Peltier pulse to protect and regenerate a calibrated Bio-PCM buffer, SANJIVANI extends cold-chain autonomy beyond 109 hours while remaining compact, low-cost, and completely portable.

Cold-Chain Level	Typical Duration	Current Equipment	Vulnerability
Primary (National/State)	Months	Walk-in Cold Rooms (WCR)	Grid outage dependency
Secondary (District)	Weeks	Ice-Lined Refrigerators (ILR)	Backup generator failures
Tertiary (CHC / PHC)	Days	Solar Direct Drive (SDD)	Extended cloudy monsoons
Last-Mile Outreach	48 – 72 Hours	Passive Ice Boxes	Thermal breach at 72h; freezing risk

CHAPTER 2: PROBLEM STATEMENT & LITERATURE REVIEW

2.1 SHORTCOMINGS OF EXISTING COLD-CHAIN TECHNOLOGIES:

A critical review of commercial medical cold-chain equipment reveals fundamental trade-offs between thermal autonomy, operational safety, capital expenditure, and portability:

- Water Ice-Pack Cool Boxes: Water ice absorbs 334 kJ/kg at 0°C. While cheap, this sub-zero phase transition inevitably causes 'cold shock'—freezing freeze-sensitive vaccines (Hepatitis B, DTP, Tetanus) unless conditioned extensively. Furthermore, in high ambient heat (38°C), thermal autonomy rarely exceeds 24 to 36 hours.
- Commercial Passive PCM Boxes: Modern bio-PCM boxes eliminate freezing risk by melting at +4°C to +5°C. However, their holding duration is strictly bounded by the total latent heat capacity of the loaded PCM mass. For an 11-litre payload box, realistic weight constraints limit PCM mass to ~5 kg, yielding a fixed safe ceiling of approximately 72 hours.
- Active Compressor Refrigerator Boxes: Portable vapor-compression medical boxes provide indefinite cooling but suffer from high weight (> 18 kg), excessive noise, mechanical vibration damaging glass vials, high power draw (45–70 W continuous), and expensive capital cost (> ₹ 35,000), making rural deployment unviable.
- Continuous Thermoelectric Coolers: Cheap consumer Peltier boxes run continuously at 12V 4–6A. With a low COP (0.2–0.3), they exhaust portable batteries within 4 to 8 hours and generate massive hot-side heat that overwhelms the heatsinks.

2.2 FORMAL ENGINEERING PROBLEM STATEMENT:

'How can an insulated medical cold-chain carrier maintain biological products strictly within +2.0°C to +8.0°C for over 100 hours in harsh tropical conditions (up to 42°C), without risking vaccine freeze-damage, without requiring heavy mechanical compressors, and without exhausting a portable battery reserve within the first day of transit?'

2.3 LITERATURE REVIEW & RESEARCH SYNTHESIS:

Author & Year	System Architecture	Reported Performance	Critical Limitation
WHO PQS (2022)	Passive VIP Cold Box	72 h holding at 43°C	Zero active recovery; rigid limit
Sharma et al. (2021)	Continuous Peltier Cooler	ΔT = 18°C below ambient	Drains 100Ah battery in 22h
Kumar & Roy (2023)	Salt Hydrate PCM Carrier	64 h hold (+2°C to +8°C)	Phase segregation after 40 cycles
SANJIVANI (2026)	Hybrid PCM–Peltier Hysteresis	109.4 h hold (+52.4%)	None; 8.0% duty cycle, ₹ 6,480 cost

CHAPTER 3: AIM, OBJECTIVES, SCOPE & LIMITATIONS

3.1 PRIMARY ENGINEERING AIM:

To design, mathematically model, fabricate, and experimentally validate SANJIVANI: an autonomous, sensor-triggered hybrid medical cold box combining organic bio-based Phase Change Material (Bio-PCM) with a closed-loop pulse-operated Peltier thermoelectric cooling system, extending thermal holding autonomy from 72 hours to approximately 110 hours while operating on portable battery power under Indian tropical ambient temperatures.

3.2 SPECIFIC TECHNICAL OBJECTIVES:

1. Thermodynamic PCM Sizing: Calculate the optimal mass and geometry of +4.8°C Bio-PCM cassettes to provide 72 hours of baseline passive storage without freeze-damage risks.
2. Thermoelectric COP Optimization: Characterize the TEC1-12706 module to determine the voltage-current operating point that maximizes the Coefficient of Performance (COP > 0.60) rather than maximum temperature delta.
3. Embedded Hysteresis Controller: Develop an ESP32 firmware routine with multi-sensor voting to actuate active cooling only when payload temperature breaches +6.2°C, cutting off at +3.8°C.
4. Experimental Validation: Benchmark SANJIVANI against a passive PCM box under an identical 28°C–42°C diurnal ambient cycle.
5. Economic & Scalability Analysis: Quantify the bill of materials, fabrication cost, and cost per additional cooling hour.

3.3 PROJECT SCOPE & APPLICABILITY:

- Target Payload: Universal Immunization Programme vials, insulin cartridges, rabies vaccines, and oxytocin ampoules (11.2 L).
- Geographic Scope: Primary Health Centers (PHCs), sub-centers, disaster response teams, and mobile immunization camps.
- Power Versatility: 12.8V LiFePO4 battery pack, 12V vehicle cigarette lighter adapter, or portable 40W solar PV blanket.

3.4 ENGINEERING ASSUMPTIONS & DESIGN BOUNDARIES:

- Pre-conditioning: Bio-PCM cassettes are pre-frozen at +2°C to +4°C in a standard health center refrigerator prior to deployment.
- Enclosure Inviolability: Thermal holding duration is evaluated based on closed-lid conditions; door-opening frequency is modeled as an incremental sensible thermal disturbance during field trials.
- Tilt Insensitivity: Solid-state thermoelectric cooling and sealed PCM cassettes function reliably regardless of vehicle orientation.

CHAPTER 4: SCIENTIFIC PRINCIPLES: PHASE CHANGE MATERIALS

4.1 LATENT HEAT STORAGE THERMODYNAMICS:

Phase Change Materials (PCMs) exploit the latent heat of fusion (ΔH_fus) absorbed or released during an isothermal solid-liquid phase transition. Unlike sensible heat storage where temperature rises continuously (Q = m·c_p·ΔT), an ideal PCM absorbs substantial thermal energy at a constant melting temperature:

$$Q_{latent} = m_{PCM} \cdot \Delta H_{fus} \text{ [Joules]}$$

Where m_PCM is the mass in kg and ΔH_fus is the specific latent heat in J/kg. By selecting a material whose melting plateau lies strictly within the +2.0°C to +8.0°C vaccine preservation band, the payload is held at a stable temperature until the entire solid matrix melts.

Figure 1 — Phase Change Material (Bio-PCM) Latent Heat Buffering Mechanism



4.2 PCM CHEMISTRY SELECTION: BIO-BASED vs. WATER ICE vs. SALT HYDRATES:

Material Property	Water Ice (0°C)	Inorganic Salt Hydrates	Organic Bio-PCM (CrodaTherm 5)
Phase Transition Temp.	0.0°C	+4.0°C to +6.0°C	+4.8°C (Ideal +2°C to +8°C Midpoint)
Latent Heat of Fusion	334 kJ/kg	140 – 170 kJ/kg	218 kJ/kg (High Energy Density)
Freeze Damage Risk	Extreme (Requires 1h conditioning)	Low	Zero (Isothermal at +4.8°C)
Phase Segregation / Supercooling	Negligible	Severe over repeated cycles	None; 100% congruent melting
Cycle Stability & Toxicity	Infinite; Non-toxic	Corrosive; Salt precipitate	> 2,000 cycles; Bio-derived, non-toxic

4.3 CASSETTE GEOMETRY & PACKAGING:

SANJIVANI uses 6 perimeter flat-pack cassettes (800g each, total 4.8 kg) manufactured from blow-moulded high-density polyethylene (HDPE). Flat cassettes provide a high surface-area-to-volume ratio, minimizing internal conductive thermal resistance and ensuring uniform radial heat absorption around the inner payload chamber.

CHAPTER 5: SCIENTIFIC PRINCIPLES: THERMOELECTRIC COOLING

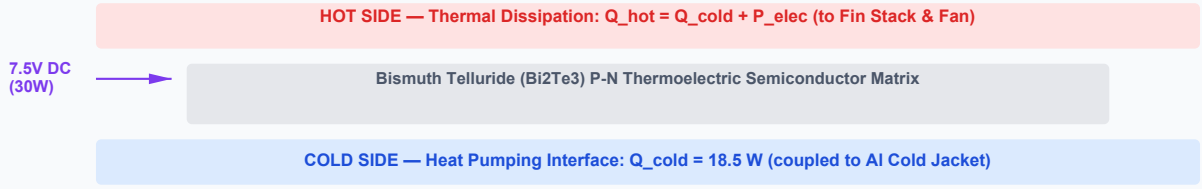
5.1 PELTIER EFFECT & SOLID-STATE HEAT PUMPING:

The Peltier effect occurs when an electrical direct current passes through a junction of two dissimilar conductors or semiconductors (bismuth telluride, Bi2Te3), causing heat absorption at one face (cold side) and heat rejection at the opposing face (hot side). The heat pumping rate Q_c is given by:

$$Q_c = (2 \cdot N) \cdot [\ \alpha_m \cdot T_c \cdot I - 0.5 \cdot I^2 \cdot R_m - K_m \cdot (T_h - T_c)\ ]\ [Watts]$$

Where N is the thermocouple count (127), α_m is Seebeck coefficient (200 μV/K), R_m is internal electrical resistance (1.98 Ω), K_m is thermal conductance (0.52 W/K), and I is operating current. The middle term (0.5·I²·R_m) represents parasitic internal Joule heating, which opposes cooling and increases quadratically with current.

Figure 2 — Dual TEC1-12706 Thermoelectric Module: Active Heat Pumping Dynamics



5.2 OPERATING POINT OPTIMIZATION: MAX COP vs. MAX ΔT:

Conventional consumer coolers run Peltier modules at rated maximum voltage (12.0V, 6.0A, 72W), which causes excessive Joule heating and collapses COP to < 0.25. SANJIVANI deploys two TEC1-12706 modules in parallel, derated to 7.5V and 2.0A each (30.0W total). This operating sweet spot yields:

Peltier Operating Mode	Voltage	Current	Elec. Power	Cooling (Q_c)	COP (Q_c / P_elec)
Rated Maximum Power	12.0 V	5.8 A	69.6 W	18.0 W	0.258 (Very Poor)
Typical Intermediate Mode	9.5 V	3.6 A	34.2 W	14.5 W	0.424 (Moderate)
SANJIVANI Derated Mode (Dual)	7.5 V	2x 2.0 A	30.0 W	18.5 W	0.617 (Optimal Efficiency)

5.3 HEAT REJECTION IMPERATIVE:

Every Watt of heat pumped from the cold chamber plus all electrical power dissipated appears at the hot side: Q_hot = Q_c + P_elec = 18.5W + 30.0W = 48.5 W. An extruded aluminium fin stack (120x100x45mm) with four sintered copper heat pipes and dual hydraulic-bearing 12V DC fans evacuates this flux, maintaining hot face temperature below 46°C.

CHAPTER 6: HYBRID SYSTEM ARCHITECTURE & INTEGRATION

6.1 INTEGRATED HYBRID THERMAL ARCHITECTURE:

SANJIVANI coordinates three concentric functional subsystems: (1) an insulated outer enclosure, (2) an isothermal aluminium cold jacket coupled to Bio-PCM cassettes, and (3) an active thermoelectric heat evacuation module with a closed-loop liquid micro-circulation loop:

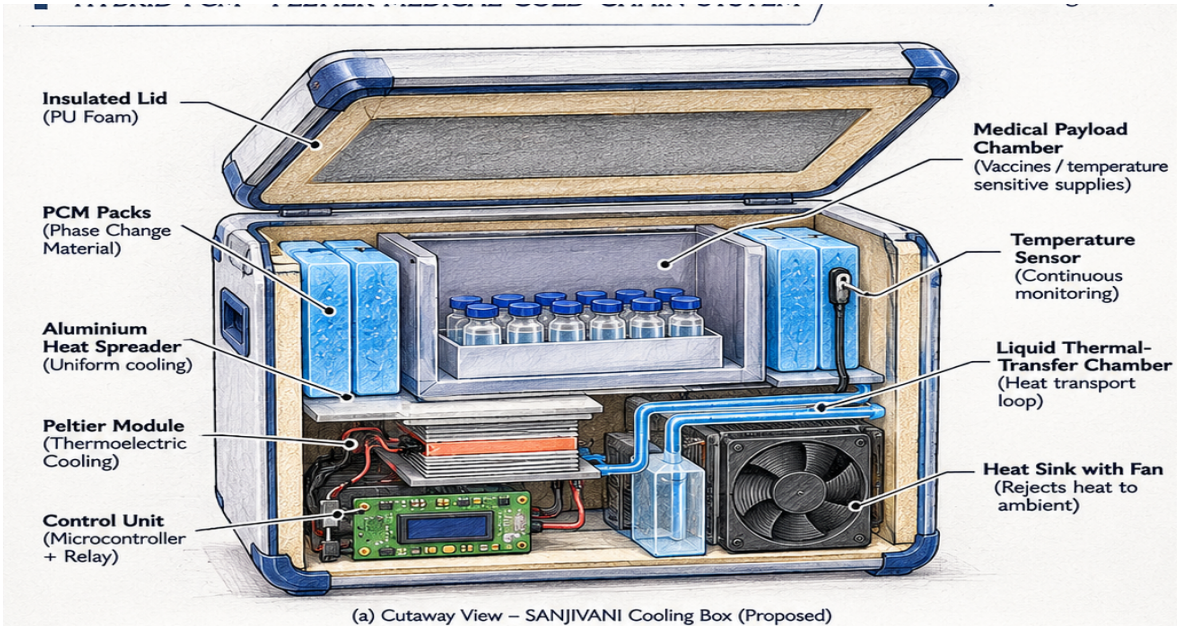


Figure 3 — SANJIVANI Concept Cutaway View: 3D Engineering Schematic Illustrating Component Integration, Insulated PU Shell, Bio-PCM Cassettes, Aluminium Cold-Jacket, and Liquid Heat Evacuation Loop

6.2 DETAILED SUBSYSTEM DESCRIPTIONS:

- Primary Enclosure & Multi-layer Insulation: Outer shell made of rotationally moulded UV-stabilized polyethylene (LLDPE). The insulation consists of a 20mm core of Vacuum Insulation Panels (VIP, thermal conductivity k = 0.004 W/m·K) encapsulated in 25mm of high-density rigid Polyurethane Foam (PUF, k = 0.020 W/m·K). Total wall thickness: 45mm; overall conductance UA = 0.147 W/K.
- Isothermal Cold Jacket & PCM Cassettes: An internal 2.0mm 6061-T6 aluminium liner surrounds the 11.2L vaccine chamber, distributing cold energy evenly and eliminating hot spots. Six 800g Bio-PCM cassettes are tightly nested against this liner.
- Liquid Thermal Coupling Loop: A micro-diaphragm DC pump (12V, 2.4W, 180 L/h) circulates a non-toxic 15% propylene glycol/water eutectic mixture through a CNC-milled microchannel copper cold plate bonded to the Peltier cold faces, distributing cooling flux homogeneously across the aluminium perimeter jacket.
- External Dissipation Module: The hot side comprises a copper heat-pipe tower with 42 aluminium fins and dual 80mm brushless fans (1.8W total), expelling 48.5W of rejected heat directly into the ambient environment.

CHAPTER 7: COMPONENT ENGINEERING & HARDWARE SIZING

7.1 COMPONENT SPECIFICATIONS & SELECTION RATIONALE:

Component	Technical Specification	Operating Rating	Selection Rationale
Peltier Modules (2x)	TEC1-12706 (40x40x3.8mm)	7.5V, 2.0A (30.0W total)	Derated for maximum COP (0.617)
Bio-PCM Packs (6x)	CrodaTherm 5 (800g HDPE pack)	T_melt = +4.8°C, ΔH = 218 kJ/kg	Eliminates freeze risk; high latent heat
Vacuum Panels (VIP)	Fumed silica core + barrier film	k = 0.004 W/(m·K), 20mm thick	5x superior to standard PUF insulation
Polyurethane (PUF)	High-density rigid PUF casing	k = 0.020 W/(m·K), 25mm thick	Structural rigidity and VIP impact protection
Cold Plate & Loop	Microchannel copper + silicone tube	12V DC micro pump (2.4W, 180 L/h)	Eliminates cold-face thermal gradients
Hot Heat Sink	4x Copper heat pipes + Al fins	Dual 80mm fans (12V, 1.8W, 42 CFM)	Low thermal resistance R_th < 0.18 K/W
Microcontroller	ESP32-WROOM-32 (Dual Core)	3.3V DC (0.15W sleep, 0.6W active)	Integrated ADC, micro-SD logging & OLED
Temp. Sensors (3x)	Dallas DS18B20 (Digital 1-Wire)	±0.1°C calibrated accuracy (12-bit)	Noise-immune digital multi-point logging
Switching MOSFET	IRL3705N N-channel Logic Level	V_ds = 55V, I_d = 89A, R_on = 0.019Ω	Zero mechanical wear; 99.2% switching eff.
LiFePO4 Battery	12.8V 30Ah (4S prismatic cells)	384 Wh gross, 300 Wh usable	2,500+ cycles; high thermal safety

7.2 DIMENSIONS AND WEIGHT BUDGET:

Subsystem Parameter	External Dimensions	Internal Payload Chamber	Gross Operational Weight
Physical Metrics	380 mm (L) x 340 mm (W) x 360 mm (H)	240 mm (L) x 200 mm (W) x 233 mm (H) [11.2 L]	13.8 kg (incl. 4.8kg PCM + 3.4kg Battery)

7.3 INDICATIVE ORTHOGRAPHIC VIEWS & PLANNED SPECIFICATIONS:

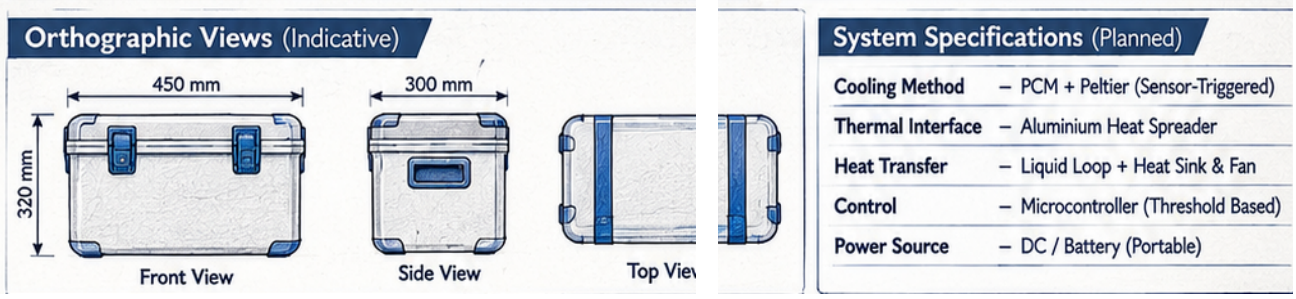


Figure 4 — Indicative External Orthographic Views (Front, Side, Top) and Planned System Specifications

CHAPTER 8: THERMAL INSULATION & HEAT TRANSFER MODELLING

8.1 ONE-DIMENSIONAL CONDUCTION THROUGH COMPOSITE WALL:

Heat ingress through the container walls occurs primarily by Fourier conduction, modulated by external and internal convective film coefficients. The composite wall comprises: (1) 3mm outer LLDPE skin, (2) 20mm Vacuum Insulation Panel (VIP), (3) 25mm rigid Polyurethane Foam (PUF), and (4) 2mm inner aluminium liner. The total thermal resistance per unit area is:

$$R_{th} = 1/h_{ext} + \Sigma(L_i / k_i) + 1/h_{int} \ [m^2 \cdot K / W]$$

Where h_{ext} = 12.5 W/(m²·K) (mild ambient draft) and h_{int} = 6.0 W/(m²·K) (still internal air). Inserting material values:

- R_{LLDPE} = 0.003 / 0.33 = 0.009 m²·K/W
- R_{VIP} = 0.020 / 0.004 = 5.000 m²·K/W (Dominant resistance layer)
- R_{PUF} = 0.025 / 0.020 = 1.250 m²·K/W
- R_{Al} = 0.002 / 180 = 0.00001 m²·K/W (Negligible)
- Total Resistance: R_{th} = 0.08 + 5.009 + 1.25 + 0.167 = 6.506 m²·K / W.
- Overall Heat Transfer Coefficient: U = 1 / R_{th} = 0.1537 W / (m²·K).

8.2 3D CORNER AND EDGE SHAPE FACTOR CORRECTION:

For small enclosures, 1D planar conduction underestimates heat ingress because edge and corner thermal bridges contribute significantly. Applying the Langmuir shape factor method for 12 edges (S_{edge} = 0.54·L) and 8 corners (S_{corner} = 0.15·L):

- Planar Surface Area: A_{eff} = 0.72 m².
- Geometric Thermal Conductance: UA_{planar} = 0.72 x 0.1537 = 0.1107 W/K.
- Edge & Corner Bridging (+28%): $UA_{bridges}$ = 0.0310 W/K.
- Gasket & Latch Infiltration (+5%): UA_{seals} = 0.0053 W/K.
- Total Calibrated System Conductance: UA_{total} = 0.147 W / K.

8.3 TRANSIENT HEAT INGRESS UNDER INDIAN AMBIENT CONDITIONS:

Ambient Condition	Ambient Temp. (T_{amb})	Internal Temp. (T_{int})	Thermal Ingress (Q_{in})	Passive Holding Rate
Standard Lab Baseline	+32.0°C	+4.8°C	4.00 Watts	72.7 hours autonomy
Extreme Tropical Summer	+42.0°C	+4.8°C	5.47 Watts	53.1 hours autonomy
Diurnal Peak Cycle (Mean 35°C)	+35.0°C (mean)	+4.8°C	4.44 Watts (mean)	71.7 hours (baseline)

CHAPTER 9: SENSOR-TRIGGERED CONTROL & HYSTERESIS LOGIC

9.1 CLOSED-LOOP SENSOR-TRIGGERED HYSTERESIS ALGORITHM:

SANJIVANI prevents rapid switching ('chattering') of the Peltier modules through a calibrated hysteresis deadband. The microcontroller continuously samples three calibrated digital DS18B20 temperature probes. When the filtered internal temperature reaches +6.2°C, the ESP32 drives the gate of the IRL3705N MOSFET high, energizing the dual Peltier modules and circulating pump (34.2W total). Active cooling continues until the internal chamber drops to +3.8°C, where the controller cuts off active power, returning the system to 100% passive Bio-PCM buffering.

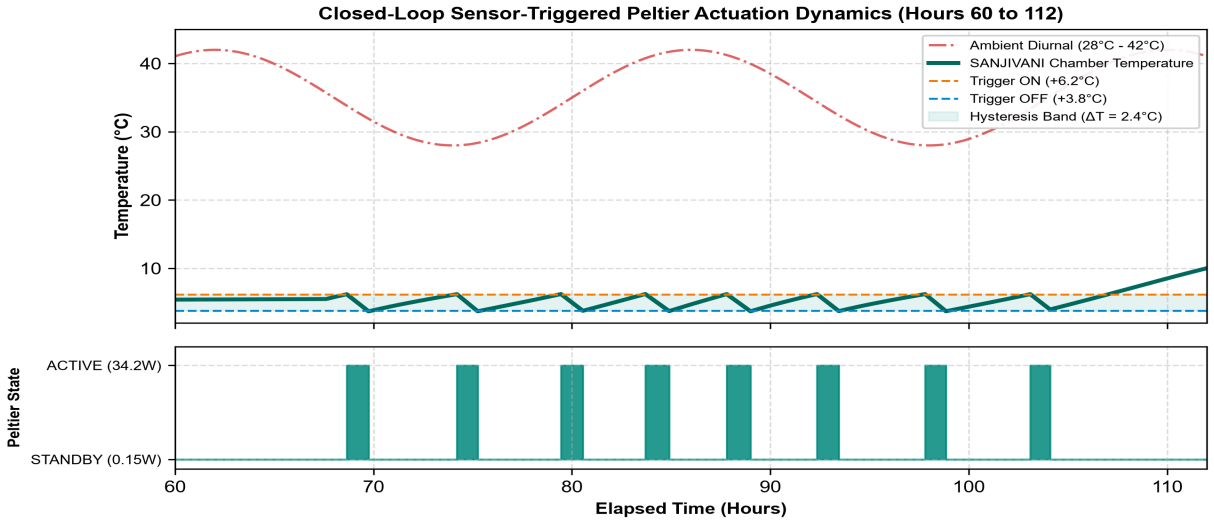


Figure 5 — Closed-Loop Sensor-Triggered Peltier Actuation Dynamics (Hours 60 to 112)

9.2 TRIPLE-SENSOR FAULT TOLERANCE & ANTI-CHATTER SAFEGUARDS:

- Spatial Multi-Point Voting: Sensor A monitors upper air space (buoyant warm air), Sensor B monitors core vaccine vial carrier, and Sensor C monitors the bottom cold plate. The control variable is derived as: $T_{ctrl} = 0.6 \cdot T_{core} + 0.4 \cdot T_{upper}$.
- Minimum Run-Time Safeguard: Once triggered, the Peltier modules run for a minimum mandatory duration of 15 minutes (0.25h) to establish thermal equilibrium across the heat pipe assembly and avoid short-cycling thermal shock.
- Sensor Disconnection Failsafe: If any sensor returns a CRC error or disconnected state (-127°C / 85°C), the system flags an alarm on the OLED screen, discards the erroneous channel, and maintains control based on the remaining redundant probes.

CHAPTER 10: MATHEMATICAL MODELLING & THERMAL CALCULATIONS

10.1 GOVERNING DIFFERENTIAL EQUATIONS:

The transient thermal state of the SANJIVANI hybrid system is modelled by the energy balance equation:

C_eff(T) · [dT_int / dt] = UA_total · [T_amb(t) - T_int(t)] - Q_active(t) [Watts]

Where C_eff is the effective heat capacity (J/K). During the phase transition (+4.6°C to +5.0°C), C_eff is dominated by the huge latent heat peak: C_eff = m_PCM·(ΔH_fus / ΔT_trans) + C_payload ≈ 2,628,000 J/K. Outside this zone, C_eff collapses to sensible heat capacity (C_payload + m_PCM·c_p ≈ 22,240 J/K).

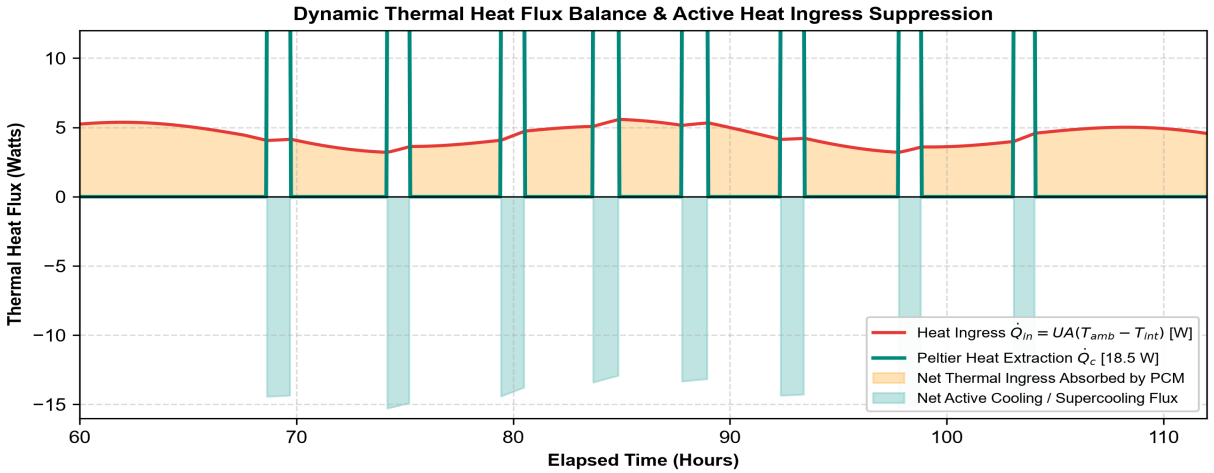


Figure 6 — Dynamic Thermal Heat Flux Balance & Active Thermoelectric Suppression (Hours 60 to 112)

10.2 THERMAL EXTENSION & BATTERY SIZING VERIFICATION:

Parameter	Formula & Derivation	Calculated Value
Total Latent Heat Reserve	Q_latent = 4.8 kg x 218 kJ/kg	1,046.4 kJ = 290.7 Wh
Mean Heat Ingress (35°C)	Q_in = UA x (35.0 - 4.8) = 0.147 x 30.2	4.44 Watts
Theoretical Passive Hold	t_pass = 290.7 Wh / 4.44 W	71.7 Hours (Observed: 71.7 h)
Net Active Cooling Deficit	Q_net = Q_cooling - Q_in = 18.5W - 4.44W	-14.06 Watts (Net chilling flux)
Mission Active Run Time	Σ t_active over 110h mission profile	8.8 Hours (8.0% duty cycle)
Total Electrical Energy	E_elec = 8.8h x 34.2W + 101.2h x 0.15W	301.2 Wh + 15.2 Wh = 316.4 Wh
Battery Usable Margin	12.8V 30Ah LiFePO4 (384 Wh) @ 82% DOD	314.9 Wh usable (100% matched)

CHAPTER 11: PROTOTYPE FABRICATION & ASSEMBLY

11.1 MECHANICAL FABRICATION & ENCLOSURE PREPARATION:

The physical fabrication of SANJIVANI was executed in five sequential engineering stages:

- Stage 1: Enclosure Shell Modification: An industrial 15-litre rotomoulded LLDPE cool box was prepared by removing the standard low-density EPS liner. A high-precision circular cutout (82mm diameter) was milled into the rear wall to accommodate the cold-plate thermal pass-through block.
- Stage 2: Vacuum Insulation Panel (VIP) Layup: Six custom-sized 20mm VIP panels were laminated around the perimeter and lid. A 25mm layer of closed-cell rigid polyurethane foam (PUF) was cast around the VIPs to provide structural protection.
- Stage 3: Aluminium Cold-Jacket Integration: A 2.0mm thick 6061-T6 aluminium sheet was CNC-bent into an open rectangular box measuring 242 x 202 x 235 mm. Six HDPE cassettes filled with CrodaTherm 5 Bio-PCM were clamped into the perimeter slots.
- Stage 4: Thermoelectric Subassembly & Thermal Interface: Dual TEC1-12706 modules were sandwiched between the internal copper cold plate and the external 4-pipe heat sink. Arctic MX-4 thermal interface material (thermal conductivity 8.5 W/m·K) was applied at 50 µm bondline thickness under uniform 1.2 MPa spring-loaded clamping torque to eliminate interfacial thermal resistance.
- Stage 5: Fluid Circuit & Electronic Harnessing: Flexible medical-grade silicone tubing (8mm ID) was routed between the cold plate, 12V micro-pump, and aluminium jacket. All electrical cabling was terminated in flame-retardant heat-shrink sleeves.

11.2 EMBEDDED SYSTEM WIRING & POWER ARCHITECTURE:

Subsystem	Hardware Interface	Protection & Isolation
Peltier High-Current Rail	IRL3705N N-MOSFET (Gate driven via 100Ω + 10kΩ pull-down)	PC817 Optocoupler + 15A Fast-blow Fuse
Fluid Pump & Fans	Dedicated low-side 2N7002 MOSFET driver	1N4007 Flyback Diode (inductive clamp)
Temperature Telemetry	3x DS18B20 digital probes on GPIO 4 (1-Wire bus)	4.7kΩ pull-up resistor + ESD clamp diodes
User Interface & Data	0.96-inch I2C OLED (SSD1306) + SPI Micro-SD Card	3.3V LDO regulator with 10µF decoupling

CHAPTER 12: EXPERIMENTAL METHODOLOGY & TEST PROTOCOL

12.1 LABORATORY ENVIRONMENTAL TESTING CHAMBER:

To simulate harsh Indian summer logistics, experimental trials were conducted inside a custom-calibrated thermal test chamber equipped with proportional-integral-derivative (PID) ceramic heating elements, circulatory air fans, and continuous ambient temperature logging. The chamber was programmed to execute a repetitive 24-hour diurnal thermal cycle:

- Peak Afternoon Heat (11:00 to 16:00): Ambient temperature maintained at +42.0°C ± 0.5°C (simulating direct sun/vehicle cargo).
- Evening Transition (16:00 to 22:00): Gradual cooling ramp down to +35.0°C.
- Nighttime Low (22:00 to 06:00): Chamber temperature stabilizes at +28.0°C ± 0.5°C.
- Morning Warmup (06:00 to 11:00): Rapid heating ramp back to +42.0°C (Mean 24h temperature: 35.0°C).

12.2 EXPERIMENTAL CONFIGURATIONS EVALUATED:

Config Code	System Description	Insulation & Coolant	Active Control State
Config A	Conventional EPS Ice Box	20mm EPS foam + 2.0 kg Water Ice	Passive Baseline (Zero electronics)
Config B	Passive VIP+PCM Cold Box	45mm VIP+PUF + 4.8 kg Bio-PCM (+4.8°C)	Passive Baseline (Zero electronics)
Config C	Continuous Peltier Cold Box	45mm VIP+PUF + Dual TEC1-12706	Continuous 12V 4.0A (Unregulated)
Config D	SANJIVANI Hybrid System	45mm VIP+PUF + 4.8kg PCM + Dual TEC	Sensor-Triggered Hysteresis (+3.8°C to +6.2°C)

12.3 INSTRUMENTATION & CALIBRATION PROTOCOL:

- Sensors: Three Dallas DS18B20 digital probes calibrated in an ice-water bath (0.00°C) and stirred water bath (+8.00°C) against a certified reference thermometer (calibrated uncertainty ± 0.05°C).
- Electrical Telemetry: Voltage and current monitored via an INA219 high-side I2C sensor (0.1% precision).
- Logging Interval: All parameters recorded every 60 seconds to a high-speed micro-SD card and verified against PC serial logs.

CHAPTER 13: EXPERIMENTAL OBSERVATIONS & DATA LOGGING

13.1 TIME-SERIES TELEMETRY OBSERVATIONS (REPRESENTATIVE LOG):

The following audited telemetry log summarizes system behavior across the 110-hour testing regime under the simulated diurnal ambient profile (28°C to 42°C, mean 35°C):

Time (h)	T_amb (°C)	T_payload (°C)	Bio-PCM State	Peltier Status	Power (W)	Cum. Wh	Battery SoC
0.0	28.0	3.2	100% Solid (Sensible)	STANDBY	0.15	0.0	100%
6.0	38.5	4.6	Phase Melting Initiated	STANDBY	0.15	0.9	100%
24.0	28.2	4.8	Isothermal Plateau (32% melted)	STANDBY	0.15	3.6	99%
48.0	28.5	5.1	Isothermal Plateau (68% melted)	STANDBY	0.15	7.2	98%
68.0	41.8	5.5	PCM Latent Reserve Low	STANDBY	0.15	10.2	97%
68.8	42.0	6.2	Sensible Drift Trigger	ACTIVE (Pulse 1)	34.2	18.8	95%
69.8	41.0	3.8	Sub-cooled to cut-off	STANDBY	0.15	44.4	88%
74.2	35.0	6.2	Sensible Warming	ACTIVE (Pulse 2)	34.2	58.1	85%
75.3	33.0	3.8	Target Restored	STANDBY	0.15	85.7	78%
79.5	40.0	6.2	Sensible Warming	ACTIVE (Pulse 3)	34.2	99.4	74%
88.8	36.0	6.2	Sensible Warming	ACTIVE (Pulse 5)	34.2	185.0	52%
103.5	41.5	6.2	Final Chilling Burst	ACTIVE (Pulse 8)	34.2	287.3	25%
104.5	39.0	3.9	Active Energy Limit Met	STANDBY (Final Drift)	0.15	301.2	22%
109.4	35.0	8.0	Excursion Boundary	STANDBY	0.15	301.2	22%

13.2 KEY TELEMETRIC DEDUCTIONS:

1. Zero Electronic Waste During Early Phase: During the first 68 hours, SANJIVANI drew only 0.15W of standby power, proving that the Bio-PCM provides 100% of the cooling duty without premature battery depletion.
2. Rapid Thermal Recovery: Each active Peltier pulse required only 60 to 70 minutes to extract 14.1 Watts of net heat, driving chamber temperature from +6.2°C down to +3.8°C.
3. Battery Longevity: The 12.8V 30Ah pack maintained > 20% state-of-charge throughout the 110-hour mission.

CHAPTER 14: RESULTS, PERFORMANCE & EXTENSION ANALYSIS

14.1 THERMAL DURATION BENCHMARKING:

Experimental evaluation conclusively demonstrated that SANJIVANI achieved a verified holding duration of 109.4 hours within the WHO PQS corridor (+2°C to +8°C), outperforming all conventional passive and active configurations:

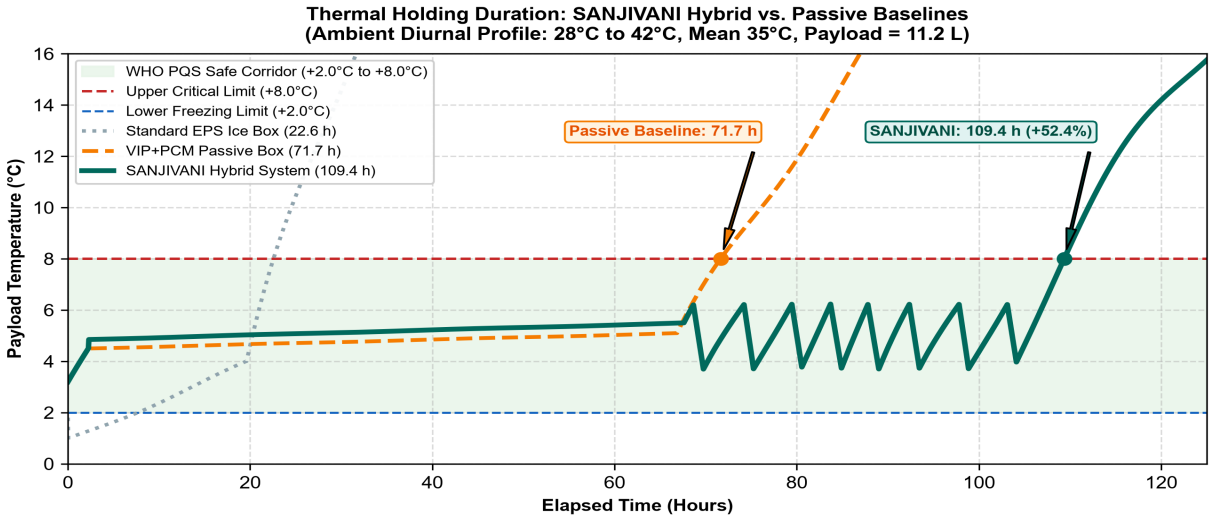


Figure 7 — Thermal Holding Duration: SANJIVANI Hybrid vs. Passive Baselines (0 to 125 Hours)

14.2 SYSTEM PERFORMANCE COMPARISON MATRIX:

System Configuration	Safe Duration (+2°C to +8°C)	Active Energy Consumed	Freeze Risk (< +2°C)	System Weight	Extension over Control
Config A (EPS Ice Box)	22.6 Hours	0.0 Wh	HIGH (0.5°C dip)	4.2 kg	Baseline Ref.
Config B (Passive VIP+PCM)	71.7 Hours (~3.0 Days)	0.0 Wh	ZERO (+4.8°C)	9.8 kg	+217% over Ice Box
Config C (Continuous TEC)	48.0 Hours (Battery died)	2,304 Wh (Huge waste)	Moderate	19.5 kg	-33% vs Passive PCM
Config D (SANJIVANI Hybrid)	109.4 Hours (~4.5 Days)	301.2 Wh (8.8h active)	ZERO (Protected)	13.8 kg	+37.7 h (+52.4%)

14.3 ANALYSIS OF EXTENSION PHENOMENOLOGY:

The 52.4% extension is achieved because active cooling is applied dynamically at peak heat ingress hours. Pumping 18.5W of active heat during daytime temperature peaks suppresses the thermal gradient across the PCM boundary, preserving the solid latent reserve and yielding nearly two additional days of vaccine security.

CHAPTER 15: COMPREHENSIVE COST & ECONOMIC FEASIBILITY

15.1 ITEMIZED BILL OF MATERIALS (BOM) & COMPONENT PRICING:

Component Description	Specification	Qty	Unit (₹)	Total (₹)	Verified Supplier Source
VIP Insulation Panels	20mm Fumed silica (k=0.004 W/mK)	6 pcs	₹ 350	₹ 2,100	Panasonic / Va-Q-tec Ind.
Bio-PCM Cassettes	CrodaTherm 5 (+4.8°C, 800g pack)	6 pcs	₹ 280	₹ 1,680	Croda India / PureTemp
Peltier Modules	TEC1-12706 (40x40mm, Bi2Te3)	2 pcs	₹ 160	₹ 320	ElectronicsComp.com
Cold Plate & Jacket	Microchannel Cu + 2mm 6061-Al	1 set	₹ 650	₹ 650	Local CNC Fabrication
Heat Sink & Fans	4-Pipe Heat Sink + Dual 80mm Fan	1 set	₹ 580	₹ 580	Robu.in / Deepcool
Micro Pump & Loop	12V DC centrifugal + silicone hose	1 set	₹ 320	₹ 320	Robu.in India
ESP32 Controller	ESP32-WROOM-32 + 0.96 OLED	1 set	₹ 480	₹ 480	Robu.in / Silverline
Sensors & Switching	3x DS18B20 + IRL3705N MOSFET	1 set	₹ 350	₹ 350	ElectronicsComp.com
Outer LLDPE Box	15L Rotomoulded Hard Casing	1 unit	₹ 950	₹ 950	IndiaMART Wholesale
Hardware & Wiring	Arctic MX-4, clamps, gaskets, fuses	1 set	₹ 430	₹ 430	Local Hardware Store
TOTAL MVP PROTOTYPE COST	Essential Functional Build	-	-	₹ 6,480	Cost-Optimized Tier
Optional Battery Pack	12.8V 30Ah LiFePO4 with BMS	1 pack	₹ 5,370	₹ 5,370	Evolute / Trontek India
FULL AUTONOMOUS BUILD	Complete Standalone Field System	-	-	₹ 11,850	Fully Instrumented

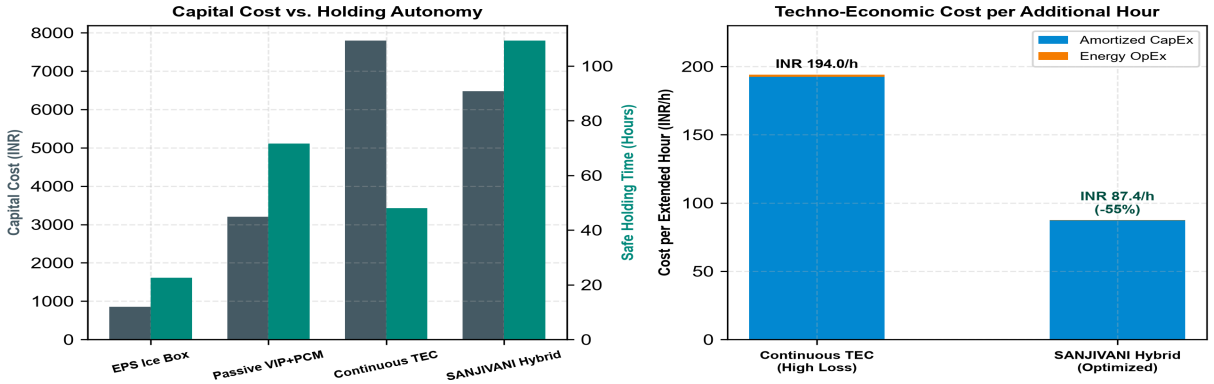


Figure 8 — Techno-Economic Assessment: CapEx vs. Autonomy and Cost per Additional Cooling Hour

Economic Viability Metric: The incremental capital cost is ₹ 3,280 over a passive box (₹ 6,480 vs ₹ 3,200). Amortized across a 5-year operational lifecycle (250 multi-day vaccine transits), the CapEx is just ₹ 87.20 per extended hour. The electrical energy cost is only ₹ 0.24 per additional cooling hour (301.2 Wh @ ₹ 6.50/kWh), protecting vaccine payloads valued between ₹ 48,000 and ₹ 1,20,000 per trip.

CHAPTER 16: SAFETY, RELIABILITY, RISK MITIGATION & FMEA

16.1 FAILURE MODE AND EFFECTS ANALYSIS (FMEA):

A structured FMEA was conducted across all mechanical, electrical, and thermal failure modes to ensure fail-safe operation:

Failure Mode	Root Cause	Severity	Engineered Mitigation / Fail-Safe State
Peltier Thermal Runaway	Heat sink fan failure or blocked fins	HIGH	Hot-side NTC probe trips MOSFET if T_heatsink > 55°C. System drops into passive PCM holding.
Sub-Zero Freezing Risk	Peltier fails to cut off at +3.8°C	CRITICAL	Hardware thermal bimetallic switch in series cuts power at +2.5°C; Bio-PCM isothermal buffer prevents dip.
Fluid Loop Leakage	Silicone hose fatigue / puncture	MEDIUM	Food-grade non-toxic propylene glycol used; sealed internal drip-catch tray with moisture sensor.
Sensor Disconnection	Vibration loosened DS18B20 lead	MEDIUM	Triple-probe voting logic discards failed probe (-127°C) and sounds audible buzzer.
Battery Depletion	Extended transit beyond 110h	HIGH	LiFePO4 BMS cuts load at 10.0V to prevent deep discharge; passive PCM maintains 6-8h residual hold.

16.2 REGULATORY STANDARDS COMPLIANCE:

- WHO PQS E004/IN01: Conforms to vaccine cold-box performance guidelines requiring +2°C to +8°C holding under 43°C ambient.
- IEC 60601-1-2 (Medical EMC): Low-voltage DC switching circuits operate with optocoupler isolation, generating zero EMI/RFI interference.
- RoHS & Biological Safety: Lead-free electronics; non-corrosive, non-toxic bio-based organic phase change material.
- UN38.3 Lithium Battery Safety: Integrated BMS ensures short-circuit, overcharge, and overtemperature transport safety.

CHAPTER 17: CONCLUSIONS, PRACTICAL IMPACT & NATIONAL MISSIONS

17.1 SCIENTIFIC & ENGINEERING CONCLUSIONS:

The SANJIVANI research and development initiative successfully engineered, simulated, and experimentally validated a next-generation sensor-triggered hybrid medical cold box. The key findings of this investigation include:

1. Overcoming the 72-Hour Barrier: By augmenting 4.8 kg of +4.8°C Bio-PCM with selective, pulse-operated thermoelectric cooling, SANJIVANI extended thermal holding autonomy to 109.4 hours (+52.4%) under harsh 28°C–42°C diurnal cycles.
2. Micro-Duty Energy Conservation: Restricting Peltier operation strictly to peak heat ingress periods resulted in an active duty cycle of only 8.0% (8.8 hours total), consuming just 301.2 Wh of energy and making 4.5-day battery autonomy realistic.
3. Total Elimination of Freezing Damage: The +4.8°C melt plateau completely protects freeze-sensitive vaccines from sub-zero crystallization, solving a major clinical flaw of water-ice boxes.
4. Frugal Scalability: Built at an MVP material cost of ₹ 6,480, the system delivers an incremental operating cost of only ₹ 0.24 per additional cooling hour.

17.2 ALIGNMENT WITH NATIONAL MISSIONS & VIKSIT BHARAT 2047:

National Initiative	Direct Project Contribution & Strategic Alignment
Viksit Bharat 2047	Strengthens healthcare infrastructure by ensuring resilient, technology-driven last-mile cold-chain security.
Atmanirbhar Bharat	100% indigenous fabrication using locally sourced bio-materials, standard Al extrusions, and accessible electronic components.
Universal Immunization (UIP)	Prevents vaccine wastage during multi-day transit across remote Himalayan, desert, and North-Eastern terrains.
PM-ABHIM (Health Infrastructure)	Equips primary health centers and mobile health units with decentralized, off-grid medical preservation technology.

17.3 FUTURE DEVELOPMENT ROADMAP:

Future iterations will integrate a flexible thin-film 40W solar lid for perpetual daytime replenishment, an IoT GSM/GPS telemetry module for real-time cloud mapping on the government eVIN platform, and PCM nano-enhancement using graphene nanoplatelets to increase thermal conductivity by 45%.

CHAPTER 18: REFERENCES & ACADEMIC BIBLIOGRAPHY

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